## Question 1: Role of PCA Before Applying the Gaussian Model

### Why PCA is used before applying the Gaussian Model

1. **Dimensionality Reduction & Computational Efficiency:**
   The credit card dataset contains 30 numerical features (V1 to V28, Amount, and Time). Calculating a full multivariate Gaussian distribution over high-dimensional space requires estimating a $d \times d$ covariance matrix ($\Sigma$) and computing its inverse ( $\Sigma^{-1}$) and determinant ($\det(\Sigma)$). Reducing dimensions via PCA significantly lowers the computational complexity from $O(d^3)$ to $O(k^3)$ where $k \ll d$.

2. **Decorrelation of Features:**
   Multivariate Gaussian anomaly detection often uses an **Independence Assumption** (or diagonal covariance matrix) to avoid calculating the full covariance matrix:

$$p(x) = \prod_{j=1}^{k} p(x_j; \mu_j, \sigma_j^2)$$

3. PCA projects the data onto orthogonal principal components, removing linear correlations between features ($Cov(Z_i, Z_j) = 0$ for $i \neq j$). This makes the feature independence assumption mathematically valid.

4. **Noise and Outlier Reduction:**
   PCA retains the axes of maximum variance (principal components) while discarding lower-variance components that often represent random noise. Eliminating noise allows the Gaussian distribution to model the true structure of normal transactions more accurately.

## 2. The Role of Threshold ($\epsilon$) in Anomaly Detection

- **What $\epsilon$ represents:** It acts as a **decision boundary on probability density**. Transactions with a likelihood below the threshold ($p(x) < \epsilon$) are flagged as anomalies (fraud), while those at or above it are classified as normal.
- **Effect of selecting different $\epsilon$ values:**
  - **High $\epsilon$ (Strict Boundary):** Increases the number of detected anomalies, increases False Positives (more false alarms), decreases False Negatives, lowers Precision, and increases Recall.
  - **Low $\epsilon$ (Permissive Boundary):** Decreases the number of detected anomalies, decreases False Positives, increases False Negatives (misses actual fraud), increases Precision, and lowers Recall. The optimal $\epsilon$ is typically chosen to maximize the $F_1$-score.

## 3. Use of a Separate Validation Set for Threshold Selection

- **Why a separate set is used:** The training set consists exclusively of normal transactions to learn baseline parameters ($\mu$ and $\Sigma$). A separate validation set containing ground-truth labels (both normal and fraud cases) is required to evaluate performance across different thresholds and select the optimal $\epsilon^*$.

- **Risks avoided:**
    - **Overfitting and Data Leakage:** Prevents tuning the threshold to the training or test data, ensuring better generalization.
    - **Distribution Contamination:** Keeps fraud cases out of the training phase so the estimated baseline parameters reflect true normal behavior.
    - **Imbalanced Metrics Bias:** Prevents evaluating performance blindly on heavily skewed data without ground-truth feedback.

## 4. Model Assumptions, Realism, and Failure Cases

- **Assumptions of the Gaussian model:**
    - Normal data follows a multivariate normal distribution ($N(\mu, \Sigma)$).
    - The training data is clean and representative of normal behavior.
    - Anomalies occur in regions of low probability density ($p(x) < \epsilon$).
    - The distribution is unimodal (forms a single central cluster).

- **Realism in fraud detection: Partially unrealistic.** Raw financial data (like transaction amounts) is heavily skewed with heavy tails rather than bell-shaped, and normal spending habits often form multiple distinct clusters (multimodal) rather than a single center.

- **When the model fails:**
    - **Mimicking Fraud:** When fraudulent transactions closely resemble normal spending patterns, placing them in high-density regions ($p(x) \geq \epsilon$).
    - **Multi-Modal Behavior:** When normal users have diverse spending habits, causing valid mid-cluster transitions to be misclassified as anomalies.
    - **Concept Drift:** When consumer behavior changes over time, causing a static model to flag legitimate seasonal shifts as anomalies.